Shiv Nadar University - Chennai

Department of Computer Science

25CS2741 Fundamentals of Internet of Things + Lab

Date: 05/10/2026 (Week 9) **Continuous Lab Evaluation-10** II Yr/ 3rd Sem B. Tech CSE(IoT)A&B

Allotted Marks: 10 (Hardware Implementation + Code Efficiency + Viva-Voce)

Topics: CubeMX + Keil MDK + STM32 Development Board + WemosD1 WiFi Module

STM32 HAL libraries are allowed

Supporting Documents: **1.** Board Schematics **2.** Reference Manual

STM32–Wemos Wi-Fi Modem Interface Using AT Commands

Problem Statement:

Interface an STM32 board with a Wemos Mini/ESP8266 through UART, configure the Wi-Fi modem using AT commands, acquire sensor data through the STM32, and transmit the sensor data over a Wi-Fi network.

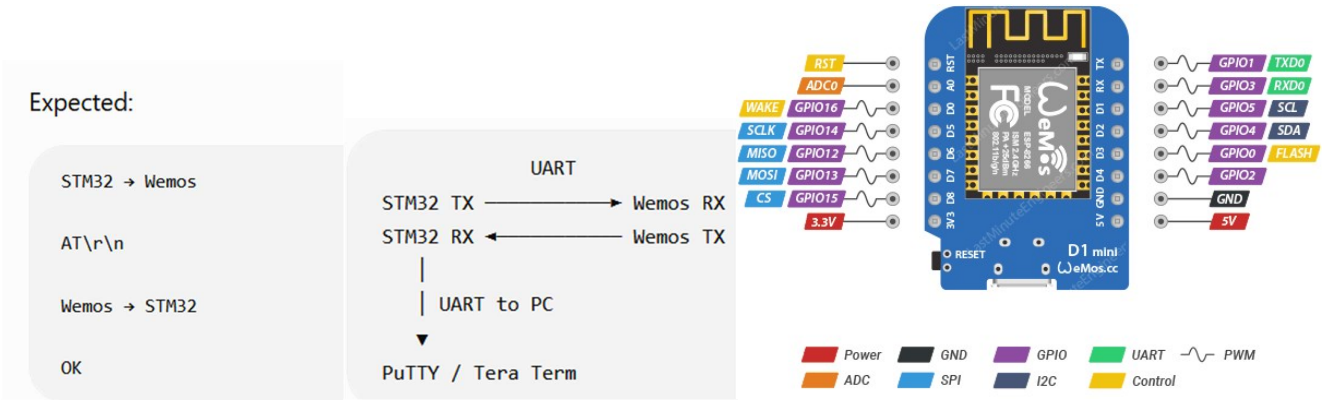
Task 9A— Establish STM32 ↔ Wemos UART Communication

Before attempting Wi-Fi, students must prove that the two boards can communicate.

Connections

Wemos D1 Mini Pin	STM32 Pin	Description
TX	**RX** (e.g., PA3 / USART2_RX)	Data from Wemos to STM32
RX	**TX** (e.g., PA2 / USART2_TX)	Data from STM32 to Wemos
GND	**GND**	Common Ground Reference
5V / 3.3V	**5V / 3.3V**	Appropriate Power Source

Configure a UART connection between STM32 and Wemos. From the STM32, transmit the command AT followed by CR/LF. Receive and display the Wemos response on the PC terminal.



Task 9B — Explore the ESP8266 AT Command Interface

Give students a **small command investigation table**, rather than a ready-made program.

Requirement	Students must determine
Check modem response	Which command?
Check firmware/version	Which command?
Set Wi-Fi station mode	Which command?
Scan available networks	Which command?

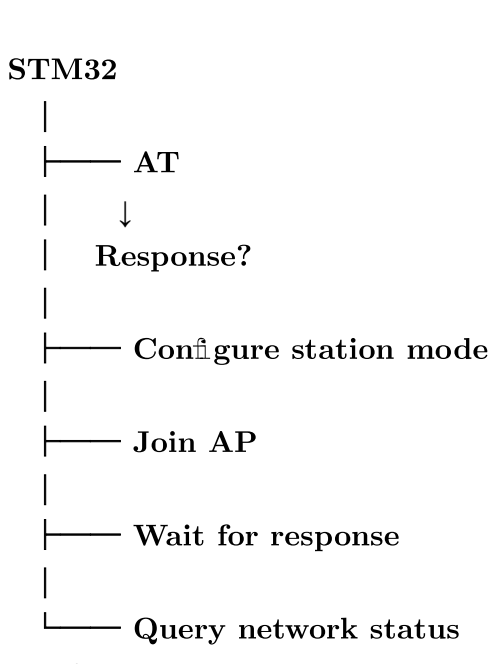
Requirement	**Students must determine**
Connect to an AP	Which command?
Obtain network information	Which command?
Establish TCP connection	Which command?
Send application data	Which command?

For example, they should discover commands in the ESP8266 AT-command documentation such as:
AT
AT+GMR
AT+CWMODE
AT+CWLAP
AT+CWJAP
AT+CIFSR

Use **two UARTs** if the STM32 board supports them:

UART	**Connection**	**Purpose**
UART1	STM32 ↔ Wemos	AT commands and responses
UART2	STM32 ↔ PC	Debug/display

Task 9C — Connect Wemos to Wi-Fi
Develop STM32 firmware that configures the Wemos ESP8266 as a Wi-Fi station and connects it to the university/local Wi-Fi network using AT commands.



Verification
STM32 should report through the PC terminal:
WIFI MODE : STATION
WIFI STATUS : CONNECTED
IP ADDRESS : xxx.xxx.xxx.xxx

STM32 sends: AT\r\n
Wemos replies: OK\r\n
STM32 receives the characters through its UART RX:
'O' → 'K' → '\r' → '\n'
STM32 then forwards the received characters to the PC terminal.
The student sees:
STM32-WiFi MODEM TEST
>> AT
<< OK

Then:
>> AT+CWMODE=1
<< OK
and:
>> AT+CWJAP="MyWiFi","password"

<< WIFI CONNECTED
<< WIFI GOT IP
<< OK